Infinite Series 6

Siddharth K

February 2026

Problem 1

$$\prod_{n=2}^{\infty} \frac{n^2-1}{n^2} = \prod_{n=2}^{\infty} \frac{n-1}{n} \prod_{n=2}^{\infty} \frac{n+1}{n} = 1 \cdot \frac{1}{2} = \frac{1}{2}.$$

Problem 2

Let $a_0 = -1$ and $a_n = 1/2^n$ for $n > 0$. The product is 0, which (technically...) diverges while the sum is $-1 + 1 = 0$, which converges.

Problem 3

Consider the sequence where $a_{2n-2} = \frac{1}{\sqrt{n}}$ and $a_{2n-1} = -\frac{1}{\sqrt{n}+1}$. The sum diverges because

$$\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n}+1} \right) = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(\sqrt{n}+1)} \leq \sum_{n=1}^{\infty} \frac{1}{n} = \infty,$$

and the product converges because

$$\left(1 + \frac{1}{\sqrt{n}}\right)\left(1 - \frac{1}{\sqrt{n}+1}\right) = 1 + \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n}+1} - \frac{1}{\sqrt{n}(\sqrt{n}+1)} = 1,$$

which means all terms of the product are equal to 1.

Problem 5

Let x_n and y_n be the width and height of the total rectangle when the $2n-1$'st and $2n$'th blocks are added respectively. We can create a recurrence. Given a rectangle made of $2n-3$ blocks, we add a rectangle to the top which makes the new area $2n-2$, so

$$y_n = \frac{2n-2}{x_{n-1}}.$$

Similarly, adding a rectangle to the side makes the new area $2n-1$

$$x_n = \frac{2n-1}{y_n} = \frac{2n-1}{2n-2} x_{n-1},$$

so

$$y_n = \frac{2n-2}{2n-3} y_{n-1}.$$

So the ratio of side lengths, $r_n = y_n/x_n$ is

$$r_n = \frac{2n-2}{2n-3} \cdot \frac{2n-2}{2n-1} r_{n-1},$$

$$r = r_1 \cdot \prod_{n=2}^{\infty} \frac{(2n-2)^2}{(2n-1)(2n-3)}.$$

From the chapter, the product is equal to $\pi/2$, so for an initial rectangle with a height-to-width ratio of r_1, the limiting ratio of side lengths is

$$\frac{\pi r_1}{2}.$$

Problem 6

Consider the partial product:

$$P = \prod_{n=1}^{\infty} \frac{3^{2^{-n}} + 5^{2^{-n}}}{2} = \lim_{N\to\infty} \prod_{n=1}^{N} \frac{3^{2^{-n}} + 5^{2^{-n}}}{2} = \left(\frac{3^{2^{-1}} + 5^{2^{-1}}}{2}\right) \cdots \left(\frac{3^{2^{-N}} + 5^{2^{-N}}}{2}\right).$$

Expanding the product by summing over all partitions of $[N] = \{1, 2, \cdots, N\}$ gives us

$$P(N) = \sum_{A \subseteq [N]} \left(\prod_{n\in A} \frac{3^{2^{-n}}}{2}\right) \left(\prod_{n\in A^c} \frac{5^{2^{-n}}}{2}\right).$$

Each smaller product is equal to

$$\prod_{n\in A} \frac{3^{2^{-n}}}{2} = 2^{-|A|} \cdot 3^{\sum_{n\in A} 2^{-n}}, \quad \prod_{n\in A^c} \frac{5^{2^{-n}}}{2} = 2^{-(N-|A|)} \cdot 5^{\sum_{n\in A^c} 2^{-n}},$$

so

$$P(N) = \sum_{A\subseteq[N]} 2^{-N} \cdot 3^{\sum_{n\in A} 2^{-n}} \cdot 5^{\sum_{n\in A^c} 2^{-n}}.$$

Define $x_A = \sum_{n\in A} 2^{-n}$, so that

$$P(N) = \sum_{A\subseteq[N]} 2^{-N} \cdot 3^{x_A} \cdot 5^{1-x_A}.$$

The numbers x_A for all $A \subseteq [N]$ are all binary numbers with N digits from 0 to 1, so they are evenly spaced by 2^{-N}. We can therefore rewrite the sum as

$$P(N) = \sum_{k=0}^{2^N} 2^{-N} \cdot 3^{k/2^N} \cdot 5^{1-k/2^N}.$$

By the definition of a Riemann integral,

$$P = \lim_{N\to\infty} P(N) = \int_0^1 3^x 5^{1-x} dx.$$

Evaluating this integral, we get

$$\int_0^1 3^x 5^{1-x} dx = 5\int_0^1 \left(\frac{3}{5}\right)^x dx = -\frac{2}{\log(3/5)}.$$

Problem 9

This product can be written as

$$\prod_{n=0}^{\infty} \frac{(8n+2)(8n+4)^2(8n+6)}{(8n+1)(8n+3)(8n+5)(8n+9)} = \prod_{n=0}^{\infty} \frac{(n+\frac{1}{4})(n+\frac{1}{2})^2(n+\frac{3}{4})}{(n+\frac{1}{8})(n+\frac{3}{8})(n+\frac{5}{8})(n+\frac{7}{8})}$$

$$= \frac{64}{35}\prod_{n=1}^{\infty} \frac{(n+\frac{1}{4})(n+\frac{1}{2})^2(n+\frac{3}{4})}{(n+\frac{1}{8})(n+\frac{3}{8})(n+\frac{5}{8})(n+\frac{7}{8})}.$$

Since $\frac{1}{4} + 2\cdot\frac{1}{2} + \frac{3}{4} = \frac{1}{8} + \frac{3}{8} + \frac{5}{8} + \frac{7}{8}$, Theorem 4.5 applies, so the product is equal to

$$\frac{64}{35} \cdot \frac{\Gamma(1+\frac{1}{8})\Gamma(1+\frac{3}{8})\Gamma(1+\frac{5}{8})\Gamma(1+\frac{7}{8})}{\Gamma(1+\frac{1}{4})\Gamma(1+\frac{1}{2})^2\Gamma(1+\frac{3}{4})}.$$

Using the recurrence $\Gamma(z+1) = z\Gamma(z)$, the product is

$$\frac{64}{35} \cdot \frac{\frac{1}{8}\cdot\frac{3}{8}\cdot\frac{5}{8}\cdot\frac{7}{8}}{\frac{1}{4}\cdot\frac{1}{4}\cdot\frac{3}{4}} \cdot \frac{\Gamma(\frac{1}{8})\Gamma(\frac{3}{8})\Gamma(\frac{5}{8})\Gamma(\frac{7}{8})}{\Gamma(\frac{1}{4})\Gamma(\frac{1}{2})^2\Gamma(\frac{3}{4})}.$$

Using the reflection formula,

$$\Gamma\left(\frac{1}{8}\right)\Gamma\left(\frac{7}{8}\right) = \frac{\pi}{\sin(\frac{\pi}{8})}, \quad \Gamma\left(\frac{3}{8}\right)\Gamma\left(\frac{5}{8}\right) = \frac{\pi}{\sin(\frac{3\pi}{8})},$$

$$\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right) = \frac{\pi}{\sin(\frac{\pi}{4})}, \quad \Gamma\left(\frac{1}{2}\right)^2 = \frac{\pi}{\sin(\frac{\pi}{2})},$$

so the product becomes

$$\frac{64}{35}\cdot\frac{35}{64}\cdot\frac{\sin(\frac{\pi}{4})\sin(\frac{\pi}{2})}{\sin(\frac{\pi}{8})\sin(\frac{3\pi}{8})} = \frac{\sqrt{2}/2}{\frac{\sqrt{2-\sqrt{2}}}{2}\cdot\frac{\sqrt{2+\sqrt{2}}}{2}} = 2.$$

Problem 15

First, we change variables: let $u = xy$ and $v = x$. Then $x = v$ and $y = u/v$, so $(x, y) \in (0, 1)^2$ maps to $0 < u < v < 1$. The Jacobian is

$$\left|\frac{\partial(x, y)}{\partial(u, v)}\right| = \frac{1}{v}.$$

So the integral becomes

$$\int_0^1 \int_0^v \frac{1 - v}{(1 - u)\log u} du\, dv = \int_0^1 \frac{1}{(1 - u)\log u} \left(\int_u^1 \frac{1 - v}{v} dv\right) du.$$

The inner integral is

$$\int_u^1 \left(\frac{1}{v} - 1\right) dv = [\log v - v]_u^1 = -1 - \log u + u.$$

So the whole integral is

$$\int_0^1 \frac{-1 - \ln u + u}{(1 - u)\ln u} du.$$

The numerator can be rewritten as $-(1 - u) - \ln u$, so the integral becomes

$$\int_0^1 \left(-\frac{1}{\ln u} - \frac{1}{1 - u}\right) du.$$

It remains to show that

$$\gamma = \int_0^1 \left(\frac{1}{\ln u} + \frac{1}{1 - u}\right) du,$$

but I couldn't figure out how to do this.